Robust keyword spotting from scratch

A leakage-safe study of audio representations, compact neural networks, and failure modes.

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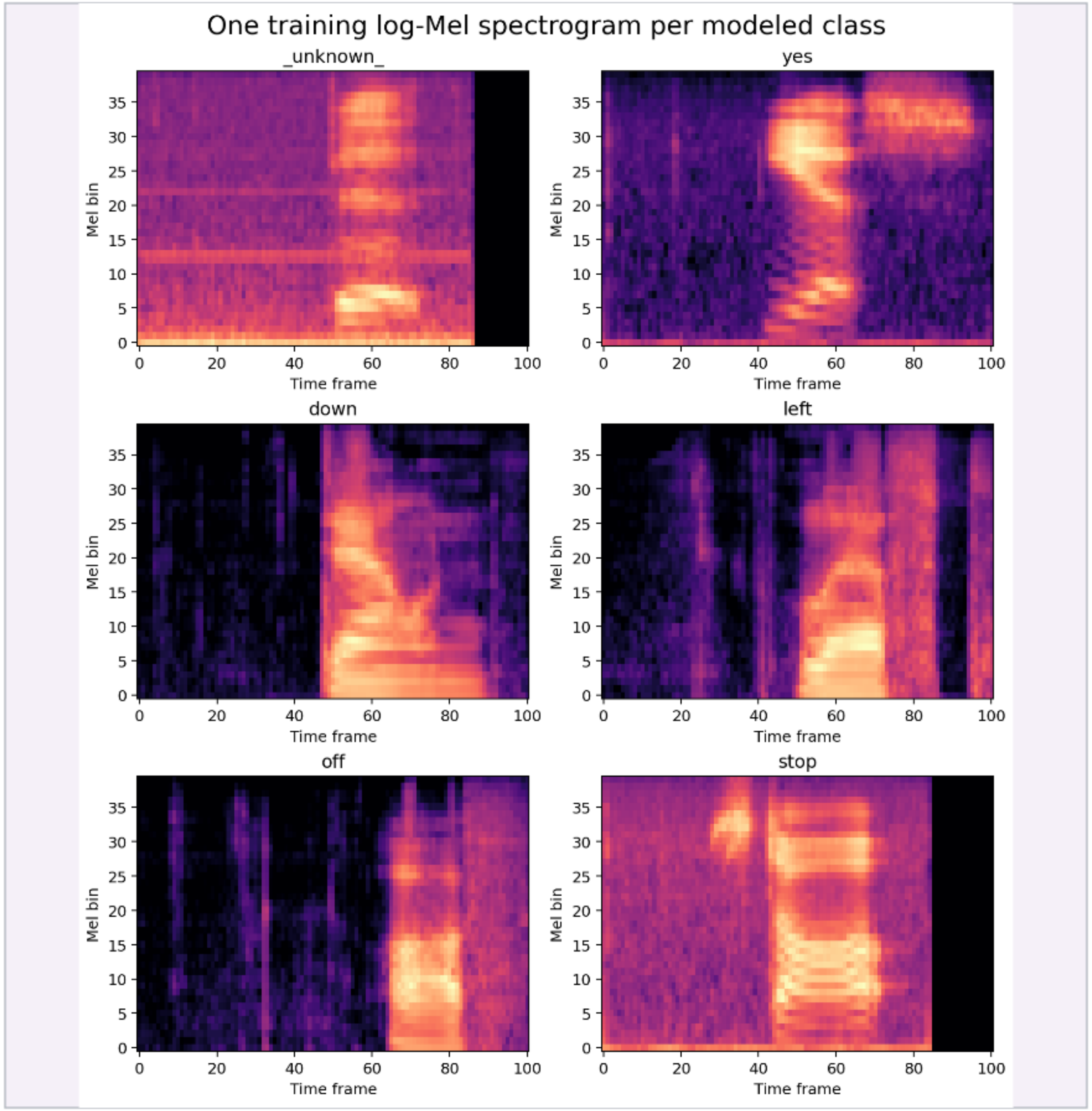


One second of audio becomes one reliable intent

Input

A mono waveform sampled at 16 kHz

Output

10 commands + unknown + silence

1 s

maximum clip duration

12

modeled classes

0

pretrained components

Educational goal: isolate the impact of representation, architecture, and augmentation without transfer learning.

Near-balanced modeling preserves the raw audit

64,721

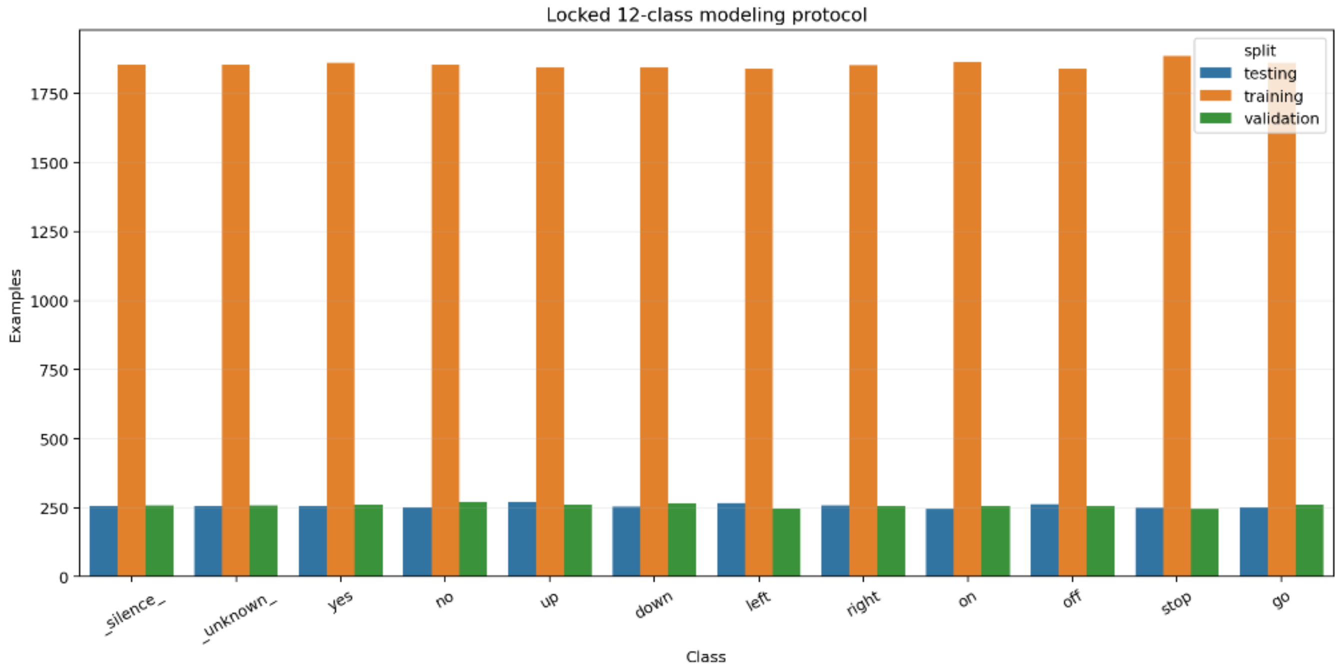
spoken clips in the raw audit

1,881

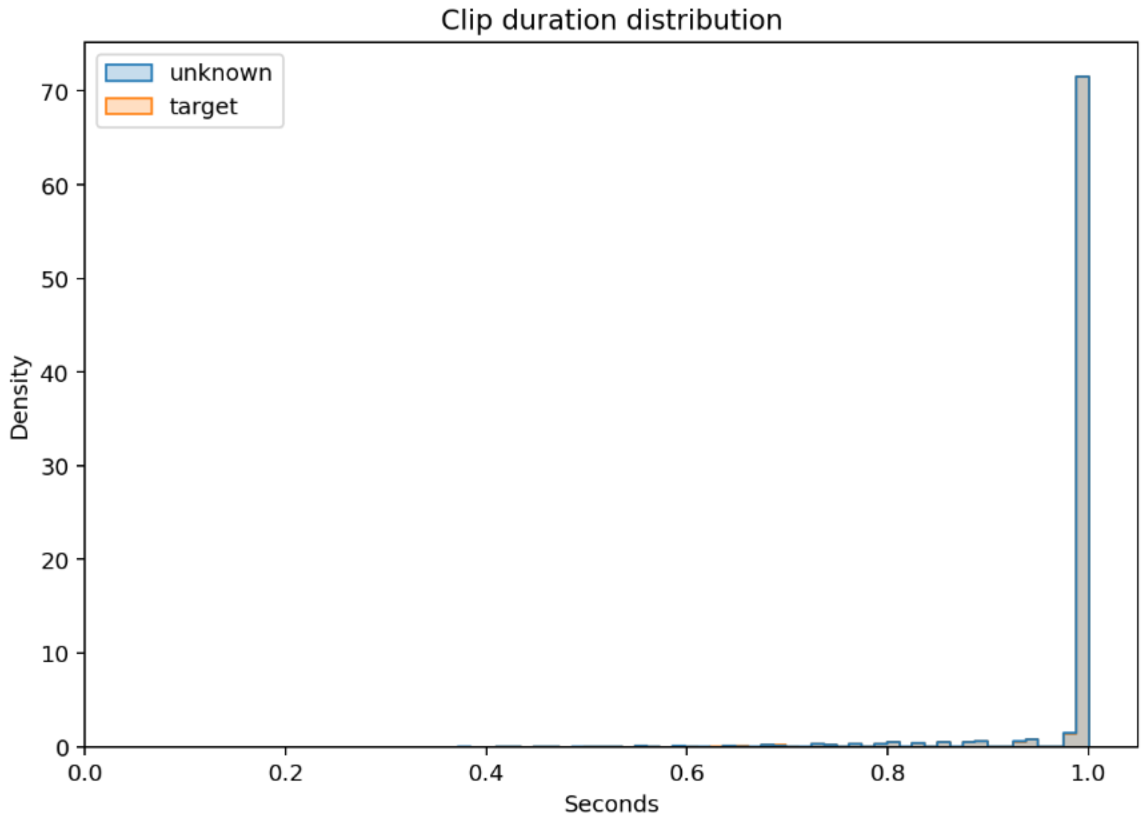
anonymized speakers

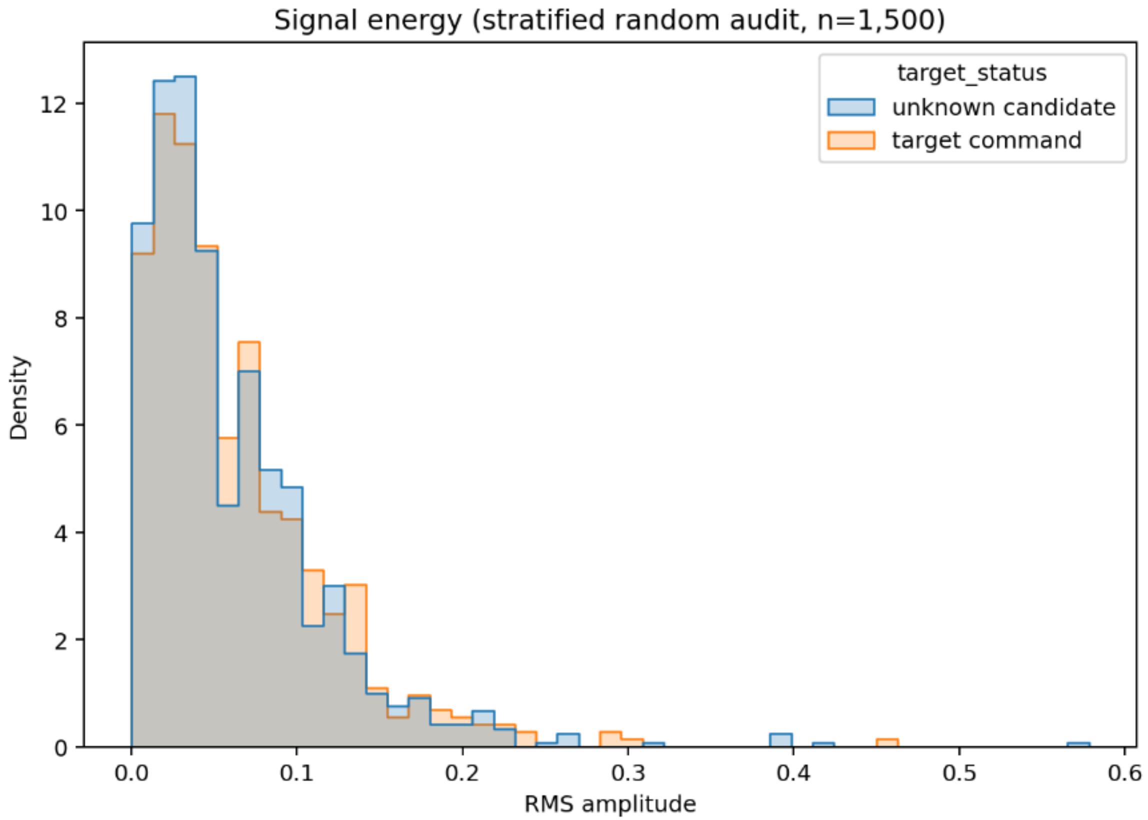
28,420 modeled examples

All target recordings are retained; unknown and silence are constructed at controlled ratios within each official split.



Clip length is stable; signal energy is not





EDA implication

Pad or trim to 16,000 samples; normalize on training data only; augment amplitude, time position, and background noise.

Speaker identity never crosses a split

Official lists define validation and test; all remaining speakers train. Background audio is reserved 80% / 10% / 10% by split.

22,246
training examples

3,093
validation examples

3,081
testing examples

Hard guards

- speaker-disjoint splits + silence regions
- split-aware noise + train-only statistics

Fail fast
on any overlap

Log-Mel preserves time-frequency structure

MFCCs are the compact baseline; main experiments retain the full log-Mel map.

Five experiments isolate each design choice

Selection metric: validation macro-F1; the test set remains sealed.

ID	Representation	Architecture	Augmentation	Question
E01	13 MFCC	MLP	None	Can a dense baseline suffice?
E02	40 log-Mel	Small CNN	None	Does local structure help?
E03	40 log-Mel	DS-CNN	None	Can efficiency retain accuracy?
E04	40 log-Mel	DS-CNN	Wave + SpecAugment	What does augmentation trade?
E05	40 log-Mel	Conv + BiGRU	Wave + SpecAugment	Does temporal memory help?

Task structure beats parameter count

MFCC MLP

361,999 params

Flattened coefficients
3 dense blocks
No explicit locality

DS-CNN

46,863 params

Depthwise + pointwise conv
Global pooling
Compact spatial model

Conv + BiGRU

292,271 params

Convolutional front end
Bidirectional recurrence
Temporal pooling

Inductive bias beats raw parameter count.

The test set stays sealed until selection is frozen

Best-model validation macro-F1 across seeds

0.9394 +/- 0.0005

The stability check did not reopen architecture selection.

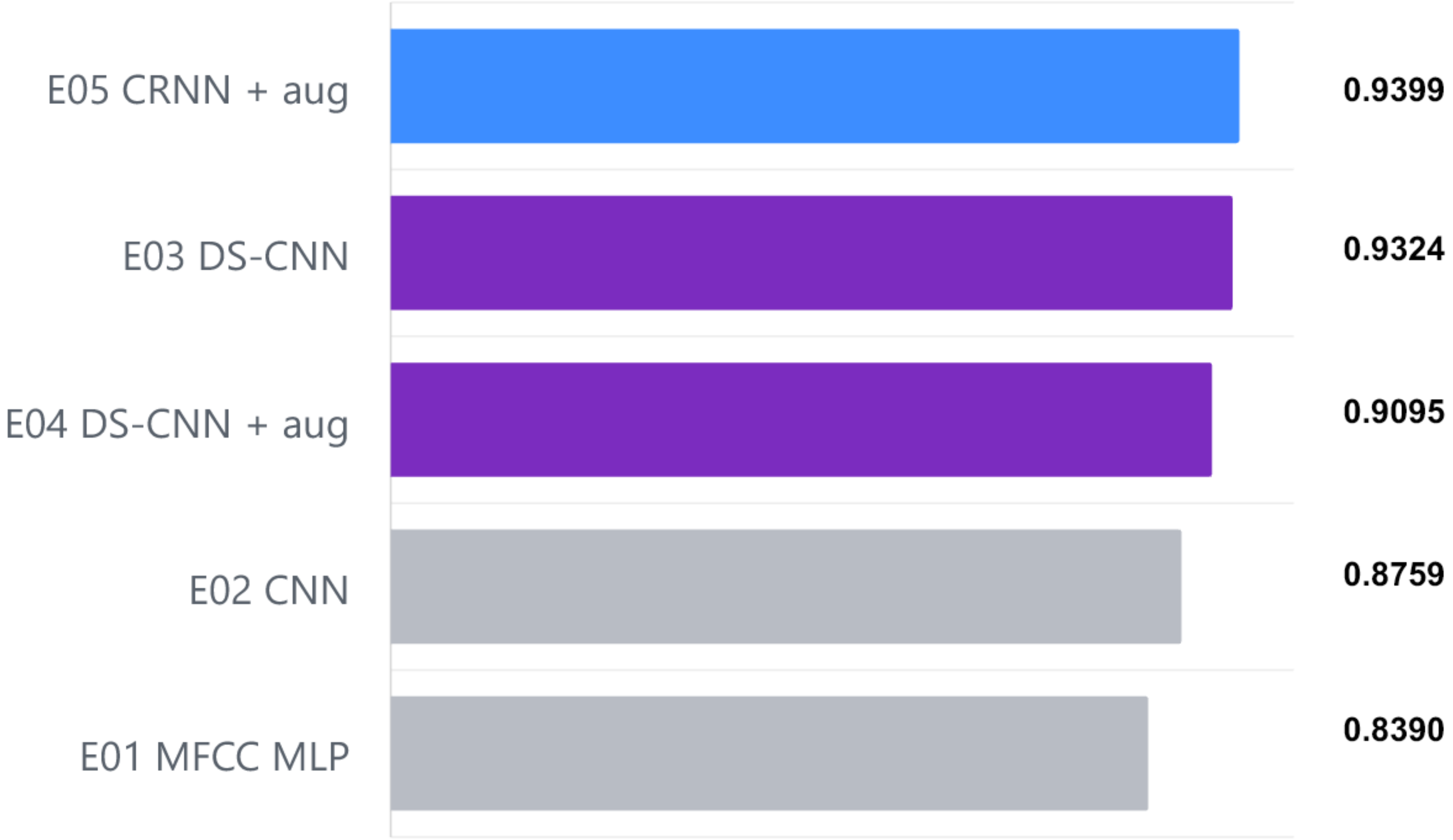
1. **Train**
   Adam + early stopping
   fixed seeds

2. **Select**
   validation macro-F1
   then accuracy

3. **Repeat**
   best config
   3 independent seeds

4. **Freeze + test**
   one clean evaluation
   then diagnostics

CRNN leads validation; DS-CNN stays compact

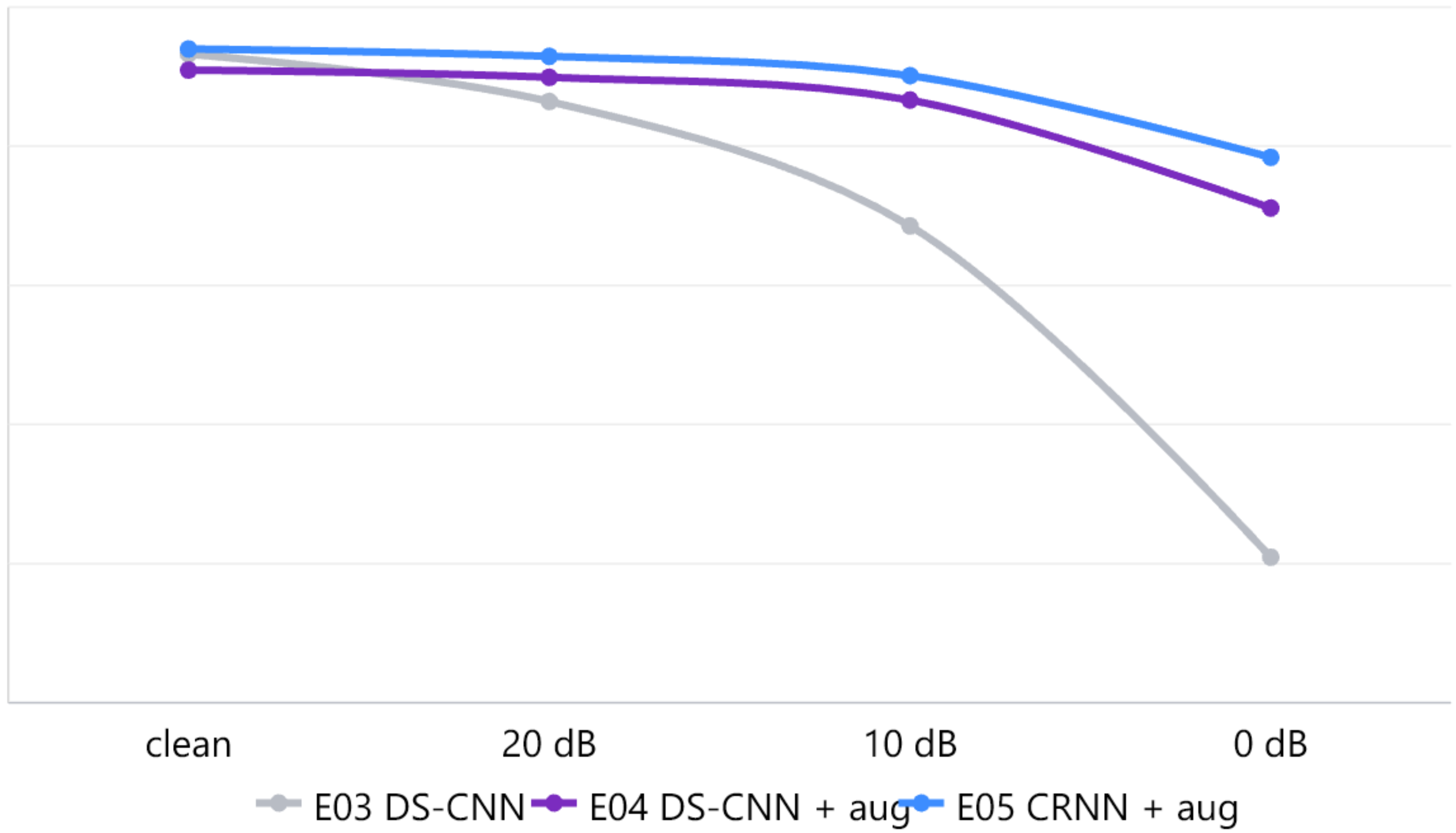


0.9399

validation macro-F1

+0.76 points over clean DS-CNN
6.24x more parameters

Augmentation buys noise robustness



-2.29 pt

clean cost for DS-CNN augmentation

+50.20 pt

0 dB gain for the same model

Frozen test confirms the validation result

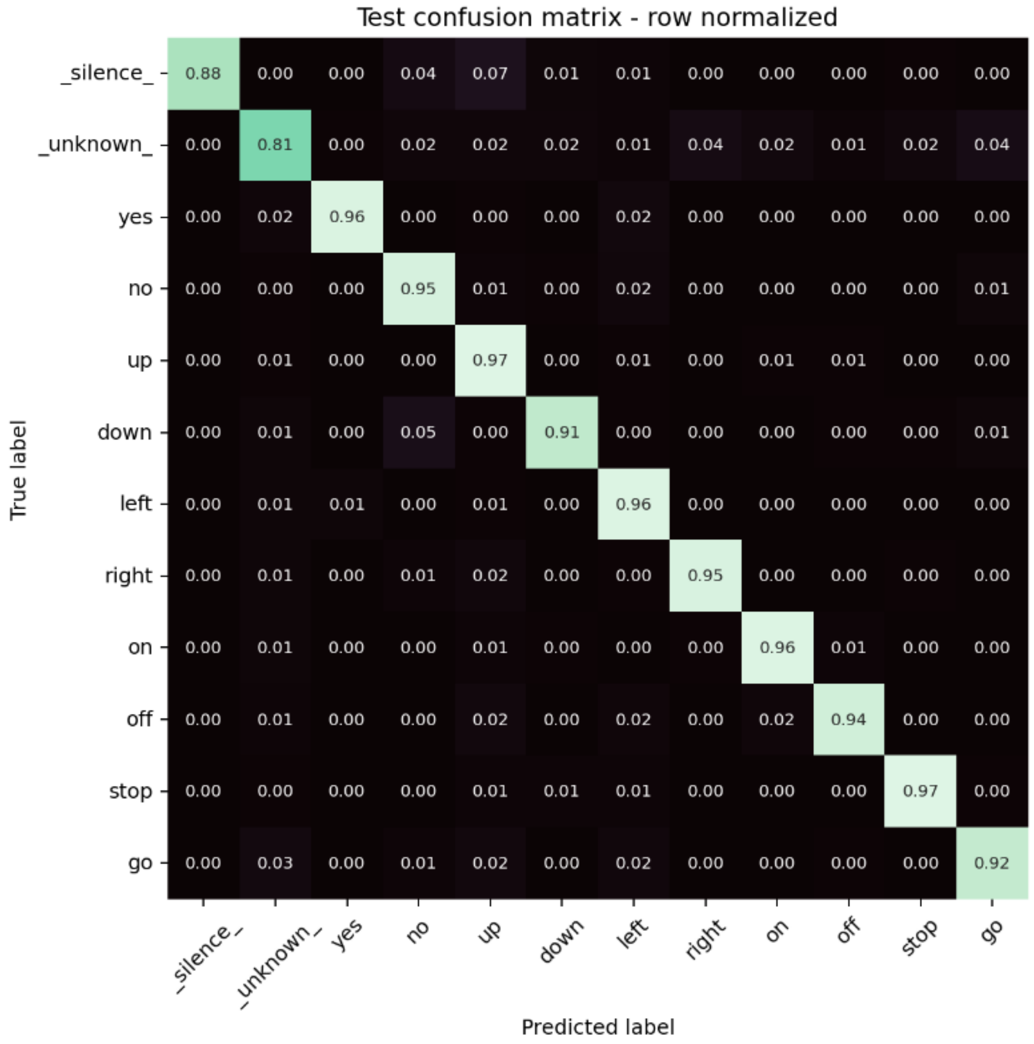
93.12%

test accuracy
95% CI: 92.24-94.00

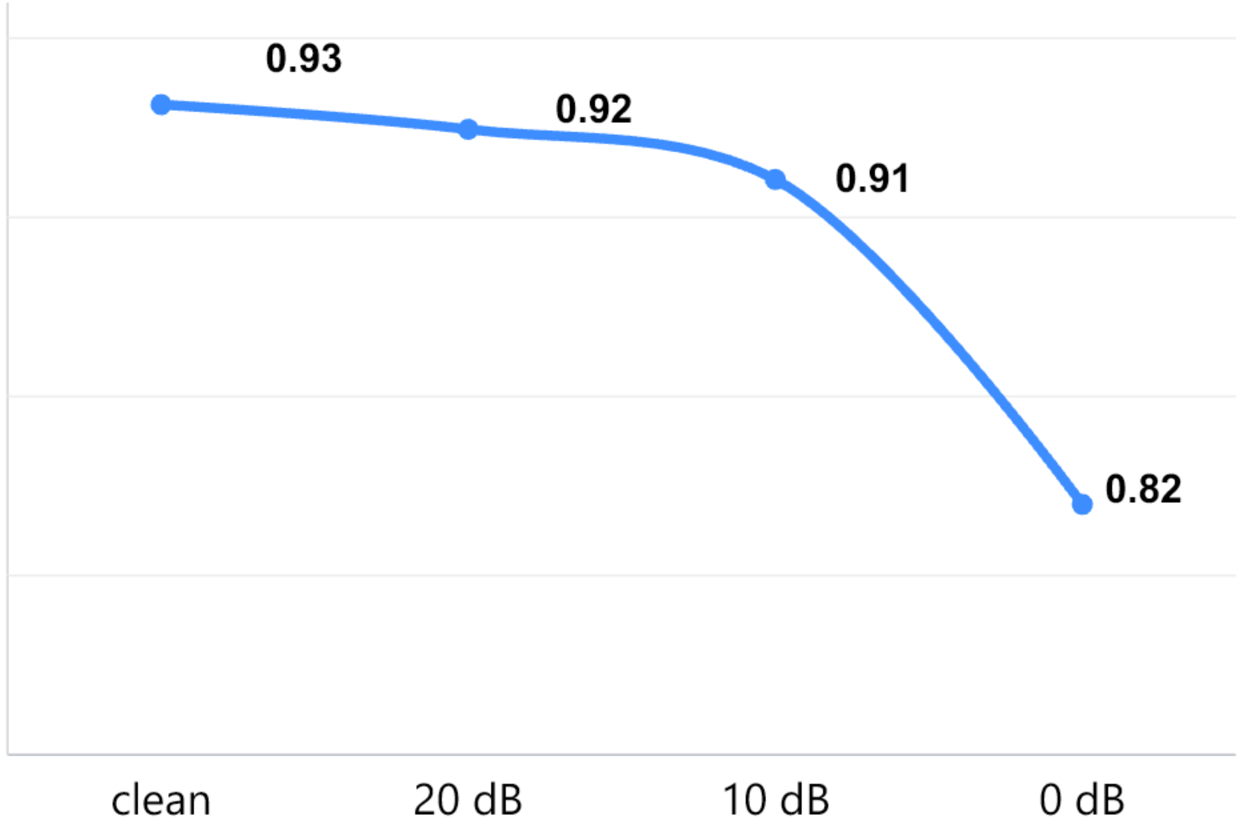
0.9315

test macro-F1
95% CI: 0.9228-0.9402

Validation -> test: -0.85 points



Noise remains the main failure axis



0.0303

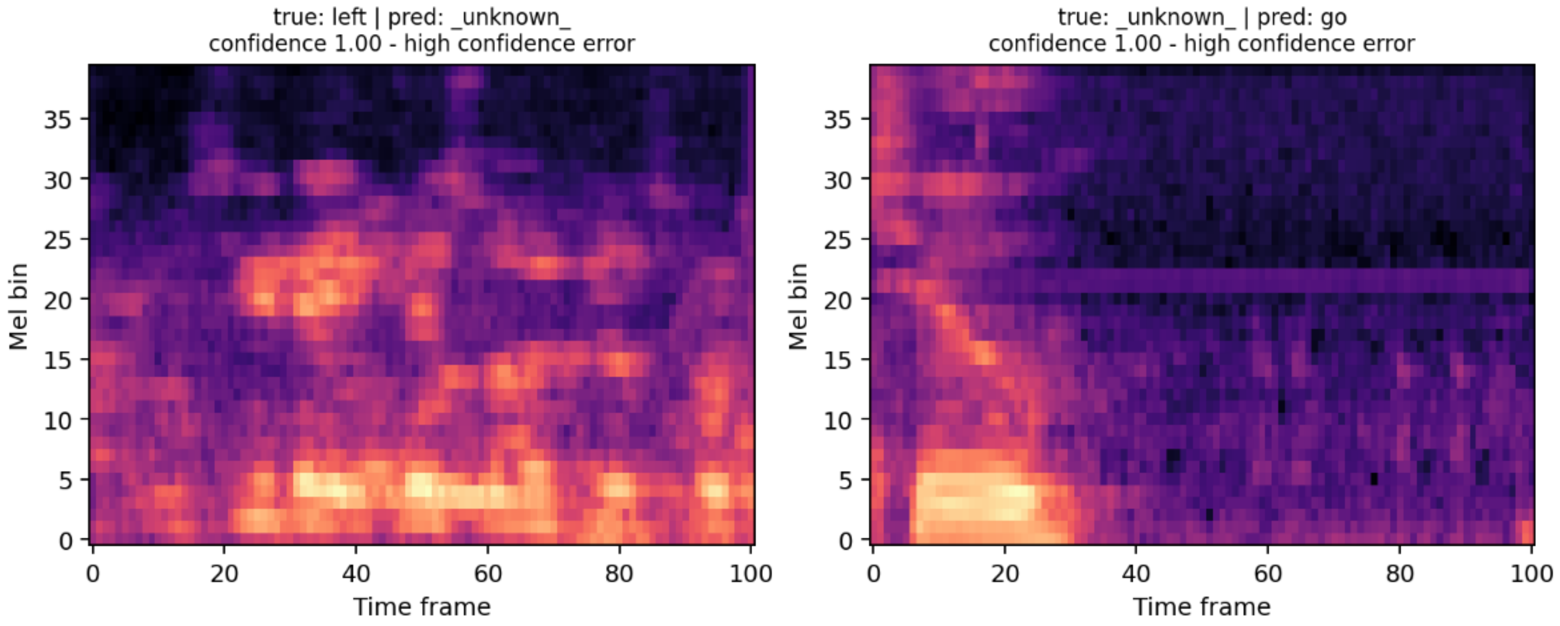
expected calibration error

3.33 ms

median CPU end-to-end latency

3.42 MiB serialized | +/-100 ms shift costs at most 0.38 points

Rare confident failures survive good calibration



Every gallery item links to a playable WAV for listening during the defense.

Weakest class

unknown F1 = 0.840

Most common error

silence -> up (17 clips)

6 confident errors

all confidence >= 0.998

Protocol matters as much as architecture

01	**Structure**	Log-Mel convolution plus temporal recurrence reached 0.9315 test macro-F1.
02	**Trade-offs**	A 46.9k-parameter DS-CNN remains compelling when footprint matters.
03	**Honest limits**	Noise, unknown speech, and rare overconfidence remain the deployment risks.

Reproducible: manifest, five runs, frozen weights, bootstrap, WAV gallery, CLI demo.